SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets
Assessment Tool Weightage: 35%

QUESTIONS: 05 **Total Marks:50**

Annexure A

Pseudocode Writing Task (Algorithm Design)

Criteria	Problem Understanding *(2)*	Algorithm Design *(3)*	Use of Control Structures *(2)*	Pseudocode Structure *(1)*	Completeness *(2)*	***Total(10)***
Weightage	*20%*	30%	20%	10%	20%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I M. Farah Afrin /192511434 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Farah Afrin

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Annexure A

Q1. Dataset: Online Shopping Transactions

Order ID	Products Purchased
O1	Mobile, Earphones
O2	Mobile, Charger
O3	Earphones, Charger
O4	Mobile, Earphones, Charger
O5	Mobile, Earphones

Question:
Design pseudocode for the **Apriori algorithm** to discover frequent product combinations from the online shopping dataset using a specified minimum support threshold. Clearly illustrate the candidate itemset generation and pruning process at each iteration. *(BL3 – 1.05 Marks)*

Q2. Dataset: Library Book Borrowing Records

Borrow ID	Books Borrowed
B1	Data Science, Python
B2	Python, Machine Learning
B3	Data Science, Machine Learning
B4	Data Science, Python, Machine Learning
B5	Python, Machine Learning

Question:
Develop pseudocode for the **FP-Growth algorithm** to identify frequent borrowing patterns without generating candidate itemsets. Explain the steps involved in constructing the FP-Tree using the given borrowing records. *(BL4 – 1.05 Marks)*

Q3. Dataset: Frequently Purchased Services

Service Combination	Support Count
{Internet}	4
{Streaming}	4
{Cloud Storage}	3

{Internet, Streaming}	3
{Streaming, Cloud Storage}	2

Question:
Write pseudocode to generate **association rules** from the given frequent service combinations. Include the procedures for confidence calculation, rule generation, and filtering based on a minimum confidence threshold. *(BL3 – 1.05 Marks)*

Q4. Dataset: Student Course Enrollment

Student ID	Courses Enrolled
S1	Python, Database
S2	Python, AI
S3	Database, AI
S4	Python, Database, AI
S5	Python

Question:
Design pseudocode for **constraint-based frequent itemset mining** where only itemsets containing the course **"Python"**are considered during the mining process. Explain how the constraint affects candidate generation and pruning. *(BL4 – 1.05 Marks)*

Q5. Dataset: Hospital Treatment Hierarchy

Patient ID	Treatments Received
P1	Healthcare, Cardiology, ECG
P2	Healthcare, Orthopedics
P3	Healthcare, Cardiology
P4	Healthcare, Cardiology, ECG
P5	Healthcare, Orthopedics

Question:
Develop pseudocode for **multilevel association rule mining** using the treatment hierarchy provided in the dataset. Explain how rules are generated at different hierarchy levels and how concept hierarchies improve pattern discovery.

Q1. APRIORI ALGORITHM

Pseudocode:

START

Input transaction dataset

Set Minimum support

Find all individual items

Count support of each item

Remove items below minimum support

Keep the remaining items as frequent itemsets

WHILE frequent itemsets are available

    Generate candidate itemsets

    Count support of each candidate

    Remove candidates below minimum support

    Keep the remaining candidate as frequent itemsets

END WHILE

Display all frequent itemsets

STOP

Candidate Generation

From the dataset:

1. itemsets:

    {mobile}

    {Earphones}

    {charger}

Generate :

2 itemsets :

{Mobile, Earphones}

{Mobile, Charger}

{Earphones, Charger}

If the support is sufficient, generate :

3. itemsets :

{Mobile, Earphone, Charger}

Q2. FP- Growth Algorithm

START

Input borrowing transactions

Set minimum support

Count the frequency of each book

Remove books below minimum support

Sort remaining books by frequency

Create the FP Tree

Create the Header Table

For each book in the Header Table

    Find its conditional pattern base

    Create a conditional FP Tree

    Generate frequent itemsets

END FOR

Display all frequent itemsets

STOP

FP-Tree construction

First count the books:

Python → 4

Machine Learning → 4

Data Science → 3

Then arrange the books according to their frequency and insert each transaction into the FP-Tree

For example:

Root
- Python
  - Machine Learning
- Data Science
  - Machine Learning

Q3. Association Rule Generation

Pseudo code :

```
START
Input frequent itemsets
Set minimum confidence
For each frequent itemset
    Generate possible rules
    For each possible rules
        Calculate confidence
        If confidence >= minimum confidence
            keep the rule
        ELSE
            Remove the rule
        END IF
    END FOR
END FOR
Display Valid association rules
STOP
```

Rule Generation:

From

{Internet, streaming}

Possible rules:

Internet → streaming

streaming → Internet

From: {streaming, cloud storage}

Possible rules:

{streaming → cloud storage}

cloud storage → streaming

Confidence:

Confidence tells us how often the second service is purchased when the first service is purchased

Example:

Confidence (Internet → streaming)

= support (Internet, streaming) / support (Internet)

= 3/4

⇒ 75%

If the minimum confidence is 70%, Thus this rule is accepted

Q4. Constraint-Based frequent Itemset mining

Dataset: Student course enrollment

Pseudocode

```
START
Input student course transactions
Set Constraint = "Python"
Find all individual courses
Generate candidate itemsets
 For each candidate itemset
     IF itemset contains "Python"
         Calculate support
       IF support >= minimum support
           Keep the itemset
       ELSE
           Remove the itemset
       END IF
    ELSE
       Remove the itemset
    END FOR
    Generate larger itemsets containing "python"
    Repeat until no more itemsets can be generated
    Display frequent itemsets containing "python"
    STOP
```

Candidate generation:

from the dataset:

{Python}

{Python, Database}

{Python, AI}

{Python, Database, AI}

only itemsets containing python are considered

Q5. Multilevel Association Rule mining

Dataset:

Pseudocode:

START

Input treatment transactions

Input treatment hierarchy

Set minimum support and confidence

Find frequent treatments at the top level

Generate frequent itemsets

For each hierarchy level

    Generate frequent itemsets at that level

    Calculate support

    Remove itemsets below minimum support

Generate Association rules

Calculate confidence

Keep rules above minimum confidence

END for

Display rules from all hierarchy levels

STOP

Treatment hierarchy

- Healthcare
  - cardiology
    - ECG
  - Orthopedics

Example:

Higher level:

Healthcare → cardiology

Lower level:

cardiology → ECG